

NEZAR SHEIKHI

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Summary

Computer Engineer and Master of Science in Applied Data Science with hands-on experience spanning embedded systems, computer vision, machine learning, and large-scale data engineering. Built real-time multi-camera tracking (MTMC) systems using YOLOv11, ByteTrack, and Re-Identification models (OSNet, ResNet-50, MobileNetV2); scalable ETL/Big Data pipelines on Databricks and Apache Spark; and GDPR/privacy-by-design data pipelines. Background in embedded C/C++ and real-time systems combined with modern ML/data engineering tooling (PyTorch, TensorFlow, PySpark, Docker).

Education

Malmö University <i>Master of Science in Applied Data Science</i>	Malmö, Sweden Sep. 2024 – June 2026
Malmö University <i>Bachelor of Science in Engineering, Computer Science</i>	Malmö, Sweden Aug. 2020 – June 2024

Experience

ASAxLab (Artificial Situation Awareness X Lab) <i>Graduate Research Assistant – Master's Thesis</i>	Malmö, Sweden Sep. 2025 – June 2026
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- Conducted literature review on multi-camera tracking, object detection, and person re-identification to define thesis direction; reviewed tracking frameworks and privacy-preservation challenges in camera-based systems, evaluating Re-ID models based on OSNet, ResNet-50, and MobileNetV2.
- Developed a real-time Multi-Target Multi-Camera (MTMC) tracking system using an edge-to-center architecture to improve situational awareness.
- Implemented YOLOv11 for high-speed object detection and ByteTrack for robust tracking; fine-tuned OSNet and ResNet-50 via transfer learning for accurate global re-identification (Re-ID), including accuracy-vs-performance trade-off analysis under different frame-skipping settings.
- Engineered the system for GDPR compliance using privacy-by-design data minimization – deleting raw video at the edge and retaining only abstract feature vectors.

Axis Communications <i>Production Support – Material Handling and SMT</i>	Lund, Sweden Jan. 2025 – June 2026
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- Managed material handling, inventory control, and production preparation for hardware manufacturing of network and security devices.
- Performed precision through-hole technology (THT) soldering on specialized security and networking hardware.

Malmö University <i>Teacher Assistant (Part-time)</i>	Malmö, Sweden Sep. 2024 – Jan. 2026
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- Supervised computer science and engineering students in laboratory settings, providing technical guidance and troubleshooting support.
- Designed updated lab exercises for multiple courses to integrate modern tools and reduce reliance on legacy software.

Sandströms Center AB <i>Embedded System Developer (Trial Service)</i>	Helsingborg, Sweden Sep. 2023 – Aug. 2024
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- Supported development, coding, and troubleshooting of real-time embedded systems.

Projects

Distributed Recommendation Pipeline – PySpark, Docker, Spark SQL	2025
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- Architected a scalable ML pipeline processing approximately 20 million user ratings using Apache Spark in a Docker-containerized distributed cluster.
- Optimized data partitioning with a 200-partition remapping strategy based on movieId, eliminating expensive shuffle operations.
- Achieved 71.6% precision using tuned Decision Tree classifiers, completing training in 273.87 seconds via dynamic resource allocation.

Distributed Analytics & ETL in Databricks – PySpark, Spark SQL, Hadoop

2025

- Processed and analyzed large-scale household energy datasets on Databricks clusters using Spark SQL for high-speed transformations.
- Engineered partitioned data pipelines in HDFS, reducing query latency through optimized cluster resource allocation.

Deep Learning Stock Index Prediction – LSTM, GRU, TensorFlow

2025

- Built sequential hybrid LSTM-GRU models, achieving a significant reduction in Mean Squared Error (MSE) versus baseline models.
- Applied advanced feature engineering and normalization to technical and economic signals to optimize predictive accuracy.

Early Detection of Heart Failure Risk – Scikit-Learn, Pandas, Random Forest

2024

- Developed ML classification models analyzing patient biomarkers to identify high-risk cardiovascular indicators with high precision.

Tiny Incremental Machine Learning – C/C++, Python, Arduino

2023

- Developed an incremental learning system on Arduino to enable smart-home behavioral automation.

Technical Skills

- **Computer Vision & AI:** YOLOv11, ByteTrack, DeepSORT, Re-ID, OSNet, TensorFlow, Keras, PyTorch, Scikit-Learn, Edge Computing
- **Big Data & Platforms:** Databricks, PySpark, Hadoop, HDFS, Spark SQL, KNIME, Docker, ETL Pipelines
- **Embedded & Systems:** C/C++, Arduino, Real-Time Embedded Systems, Signal Processing
- **Programming Languages:** Python, Java, C/C++, C#, JavaScript, HTML/CSS, SQL, PostgreSQL, MySQL
- **Data Visualization:** Tableau, Matplotlib, Seaborn, Plotly
- **Tools:** Git, VS Code, IntelliJ, PyCharm, Jupyter Notebook, Flask, React, Node.js, MATLAB, SPSS

Honors, Awards & Certifications

- **Certificate of Research Merits** – Sustainable Digitalisation Research Centre, Malmö University. Awarded for research contributions to the Federated & Privacy-Preserving Machine Learning project as part of the Vertically Integrated Project (VIP) Program, including Re-ID model evaluation (OSNet, ResNet-50, MobileNetV2) and privacy-preservation analysis in camera-based tracking systems.

June 2026
- **First Place, True Challenge Hackathon (Helsingborg Mashup)** – Won a 24-hour hackathon, developing a digital solution for the “Gör Så Här” (Do Like This) challenge.

May 2018

Languages

Arabic (Native) | Swedish (Professional/Fluent) | English (Professional/Fluent) | Kurdish (Conversational)

Certifications & Licenses

Driver’s License, Category B (B-Körkort)